



AWS SageMaker AI

Building, Training & Deploying ML
Models in the Cloud

IVÁN • GEMING • ZESHUO • CARLOS • MARTA





Contents

1. Context
2. Core Components
3. Model Deployment
4. Tutorial
5. Pricing Model
6. Conclusions
7. Bibliography & Resources



1. Context: The AIOps challenge

Bringing an **ML project** from a notebook to production involves complex infrastructure **challenges**:



Provision servers



Manage containers



**Build APIs &
integrations**



Handle scaling

1. Context: What is SageMaker AI?



SageMaker provides a **fully managed ML platform** to handle the entire AI/ML workflow, in a **production-ready environment**.



Cloud & Compute

Serverless & Scalable resources



Data & Storage

S3, DynamoDB & Monitoring



Development

VS CodeStudio & ML Pipelines



Custom AI Support

Flexible support for custom models



Production Ready

Seamless API communication



1. Context: What is SageMaker AI?

AWS SageMaker provides tools for the entire ML pipeline:



1. Data Preparation

Process and clean datasets at scale before training.



2. Model Training

- Built-in algorithms
- On-demand compute (CPU/GPU)
- Auto hyperparameter tuning
- Distributed training



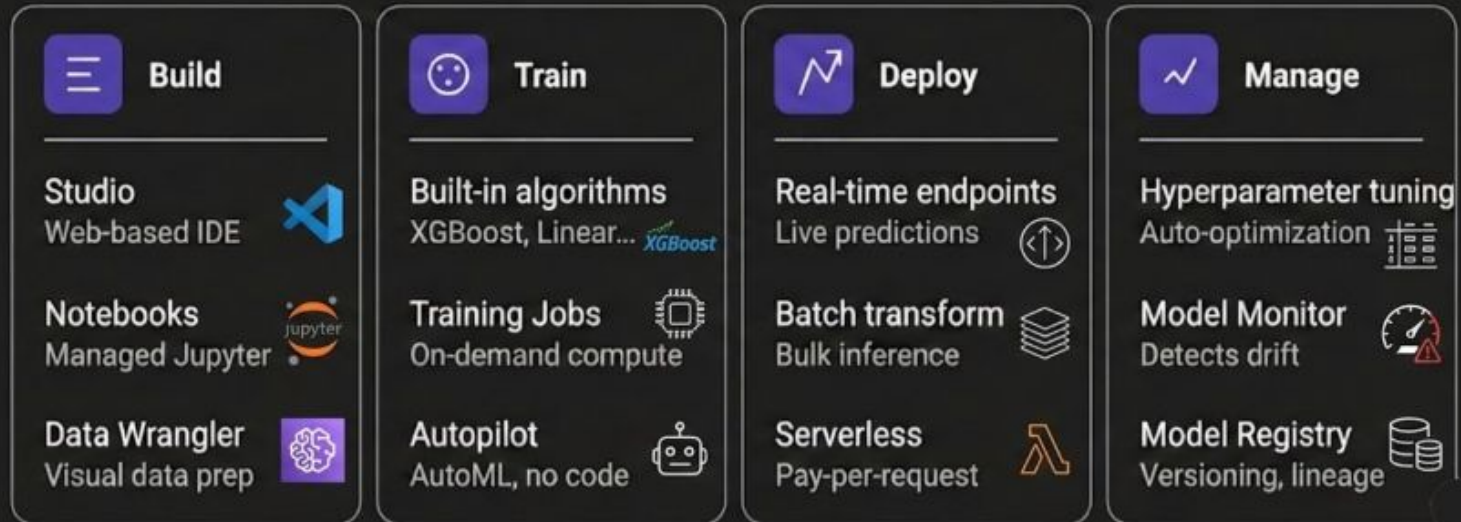
3. Deployment

Deploy models as endpoints or APIs. Receive real-time predictions and monitor performance automatically.

2. Core Components



AWS SageMaker Studio is the main interface to access all components. It provides tools for:





3. Model Deployment

SageMaker allows the **deployment of trained models** in different ways:

Real-time Inference

Used when we need immediate responses (chatbots, recommendation systems).

Batch Transform

Used for processing large datasets offline.

Serverless Inference

Automatically manages infrastructure and scaling. Pay for what we use.

Asynchronous Inference

Used when results are not needed immediately.

4. Tutorial: General overview



The screenshot shows the SageMaker Studio Home dashboard. On the left is a sidebar with navigation links: Home, Running Instances, Data, Auto ML, Experiments, Jobs, Pipelines, Models, JumpStart, and Deployments. The main area is titled 'Home' and includes an 'Overview' tab. It features two large tiles: 'JupyterLab' (orange) and 'Code Editor' (blue). Below these is a section for 'Prebuilt and automated solutions' with tiles for 'JumpStart', 'Model evaluations', and 'AutoML'. At the bottom, there are three columns of tasks: 'Prepare data', 'Build, train, tune models', and 'Deploy model'. The AWS logo is in the bottom right corner.

SageMaker Studio Provide feedback

Applications (5)

- JupyterLab
- RStudio
- Canvas
- Code Editor
- Studio CLI

Home

Launch workflows, manage your applications and spaces, and view getting started materials

Overview Getting started

Overview

Start a new ML workflow or jump back into your workflow

JupyterLab
Create, manage, and run durable instances of JupyterLab using spaces
View JupyterLab spaces >

Code Editor
Based on Code-OSS, Visual Studio Code Open Source
Create, manage, and run durable instances of Code Editor using spaces
View Code Editor spaces >

Prebuilt and automated solutions
Deploy built-in algorithms, prebuilt solutions, and example notebooks. Build models and prepare data from a visual interface.

- JumpStart**
Quickly deploy, fine-tune, and evaluate pre-trained models
- Model evaluations**
Evaluate LLMs for model quality and responsibility
- AutoML**
Automatically build, train, and tune models

Workflows and tasks

- Prepare data**
 - Connect to data sources
 - Transform, analyze, and export data
 - Store, manage, and retrieve features
 - Manage EMR clusters
- Build, train, tune models**
 - View all training jobs
 - Create an AutoML experiment
 - View all experiments
 - Tune pre-trained models
- Deploy model**
 - Get endpoint deployment recommendation
 - Deploy an endpoint in a few steps

Collapse Menu

aws

*Resources: CCDBA LUNA Project SageMaker [Tutorial](#)

4. Tutorial: JupyterLab + VSCode instances



SageMaker Studio > JupyterLab

Applications (5)

- JupyterLab
- VS Studio
- Canvas
- Code Editor
- Studio CLI

Home

Running Instances

Data

Auto ML

Experiments

Jobs

Pipelines

Models

JumpStart

Deployments

Inference recommender

Optimize cost & performance

Endpoints

Manage deployed models

Collapse Menu

JupyterLab

About

JupyterLab is the latest web-based IDE for notebooks, code, and data. You can select your image and instance to harness the power of AWS when running your machine learning applications.

See features [🔗](#) Quick start guide [🔗](#)

Filter spaces: **Running**

Name	Application	Status	Type	Last modified	Action
customer-chatbot	JupyterLab	Running	Shared	53 minutes ago	Stop Open
Datalabtest	JupyterLab	Stopped	Private	Yesterday	Run
RTC-demo	JupyterLab	Running	Shared	Yesterday	Stop Open
mlaps	JupyterLab	Stopped	Private	15 days ago	Run
JupyterAIDemoData	JupyterLab	Running	Private	16 days ago	Stop Open

13 results Results are cached Refresh Go to page 1 Page 1 of 3

Introducing spaces [New](#)

JupyterLab and Code Editor now come with durable instances that allow for faster startup, privacy options, and configurable storage.

[Learn more](#)

demo-jupyterlab-in-notebook X

Code

```
mr_ml_magics
n_sagemaker_ml_magic

Airlines-Dataset Cell Output
entastore-id Snowflake-Airlines-Da

list tables from schema 'AAAGLOBE'

MA - 'INFORMATION_SCHEMA'

ASC

S-Housing-Dataset Cell Output
entastore-id Redshift-US-Housing-Da

list columns from table 'us-housi

pg-catalog', 'pg_attr

attname atttypid attstattarget at

typname 19 -1
namespace 26 -1
typowner 23 -1
typslen 21 -1
typbtyval 16 -1
...
chunk_id 26 -1
chunk_seq 23 -1
```

Connect to cluster

Select a cluster to connect to. A code block will be added to the active cell and run automatically to establish the connection. [Learn more](#)

Name	ID	Status	Creation Time	Account ID
MyDaskProcessingClus...	j-2XZDHMOUJLYTR	Running/Waiting	2024-02-13T18:52:29	96819216650

Overview

Master nodes: 1, r5.2xlarge
Core nodes: 3, r5.2xlarge
Apps: Spark 3.4.1, hive 3.13, Livy 0.71

Application user interface
[Spark History Server](#)
[View logs](#)

Tags

for-use-with-amazon-emi-managed-policies: true

Cancel Connect

4. Tutorial: Available Endpoints



SageMaker Studio > Inference Experience > Endpoints

Provide feedback

Home

Running Instances

Data

Auto ML

Experiments

Jobs

Training

Model evaluation

Pipelines

Models

JumpStart

Deployments

Inference recommender

Endpoints

Projects

Collapse Menu

Endpoints

Endpoint are locations where you send inference requests to your deployed machine learning models. After you create an endpoint, you can add models to it, test it, and change its settings as needed.

Search by endpoint name

Delete

Create endpoint

Name	Status	Created on	Modified on
☐ xgb-model-17-04-2024-14-00-19	☒ In service	4/17/2024, 10:00:27 AM	4/17/2024, 10:02:14 AM
☐ sqlcoder-7b-deepspeed-ep-endpoint	☒ In service	4/15/2024, 6:11:12 PM	4/15/2024, 6:16:53 PM
☐ chatbot-llama2-7b	☒ In service	3/15/2024, 4:36:00 PM	3/15/2024, 4:37:47 PM

3 results

Refresh

Rows 10

Go to page 1

Page 1 of 1

Learn about endpoints

Get started

Deploy models for inference

SageMaker Inference explained: Which style is right for you?

Documentation

Real-time inference

SageMaker pricing

Deploy a Machine Learning Model to a Real-Time Inference Endpoint

What's new

Recent features launches

Engineering blog

Privacy

Site Terms

Cookie Preferences

© 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

4. Tutorial: Creating Jobs



Jupyter Lab

Code Edi...

RStudio

MLFlow

Canvas

Studio CL...

Partner AI Apps

Home

Running Instances

Compute

Data

AutoML

Experiments

Jobs

Pipelines

Models

JumpStart

Deployments

Create inference optimization job

Optimize your model for inference.

Inference optimization job

Job name*

inference-optimization-job

Optimization configurations

meta-textgeneration-llama-3-8b-045637

Optimization configurations

Deployment instance type*

inference-optimization-job

Quantization

Precision data type*

inference-optimization-job

Speculative decoding

SageMaker draft model

SageMaker provided speculative decoding solution

Use your own draft model

Bring your own speculative decoding solution

Cancel

Create project

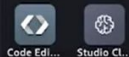
4. Tutorial: Training Jobs



SageMaker Studio > Jobs

Provide feedback

Applications (5)



Home

Running Instances

Data

Auto ML

Experiments

Jobs

Training

Model evaluation

Pipelines

Models

JumpStart

Deployments

Collapse Menu

Training Jobs

View and manage SageMaker training jobs created by notebooks, applications, and IDEs.

Search...

Filter by job type: ☒ Training ☐ Notebook jobs ☐ JumpStart training

Name	Job type	Created on	Modified on	Created by	Status
dorانبjobsplipyn-dorانبjobs-2...	Notebook training	Fri Apr 19 2024 10:21:47 GMT-04...	Fri Apr 19 2024 10:25:35 GMT-04...	genal-user	Completed
dorانبjobsplpynb-dorانبjobs-45...	Notebook training	Fri Apr 19 2024 10:16:11 GMT-040...	Fri Apr 19 2024 10:19:41 GMT-04...	genal-user	Failed
dorانبjobsplpynb-dorانبjobs-b6...	Notebook training	Fri Apr 19 2024 10:11:14 GMT-040...	Fri Apr 19 2024 10:14:48 GMT-04...	genal-user	Completed
pipelines-4jkc3qda8ekv-ModelTr...	Training	Wed Apr 17 2024 10:14:25 GMT-0...	Wed Apr 17 2024 10:16:59 GMT-0...	-	Completed
xgboost-2024-04-17-13-52-02-5...	Training	Wed Apr 17 2024 09:52:02 GMT-...	Wed Apr 17 2024 09:54:31 GMT-0...	genal-user	Completed
jumpstart-dft-meta-textgenerati...	Jumpstart training	Mon Mar 25 2024 10:20:50 GMT-...	Mon Mar 25 2024 10:41:55 GMT-...	genal-user	Completed
frauddetection-Untitled-c0cf50d...	Notebook training	Mon Mar 18 2024 11:01:26 GMT-0...	Mon Mar 18 2024 11:05:26 GMT-0...	genal-user	Completed
pipelines-ts5zn25zl8a-Abalone-T...	Training	Tue Feb 13 2024 14:53:12 GMT-0...	Tue Feb 13 2024 14:55:35 GMT-0...	-	Completed
linear-learner-2024-02-13-19-51...	Training	Tue Feb 13 2024 14:51:15 GMT-0...	Tue Feb 13 2024 14:55:08 GMT-0...	genal-user	Completed

9 results Refresh

Jobs per page 10 Go to page 1 Page 1 of 1

Learn about Jobs

Get started

- Run a SageMaker Distributed Training Job with Model

Documentation

- Train a model with Amazon SageMaker

What's new

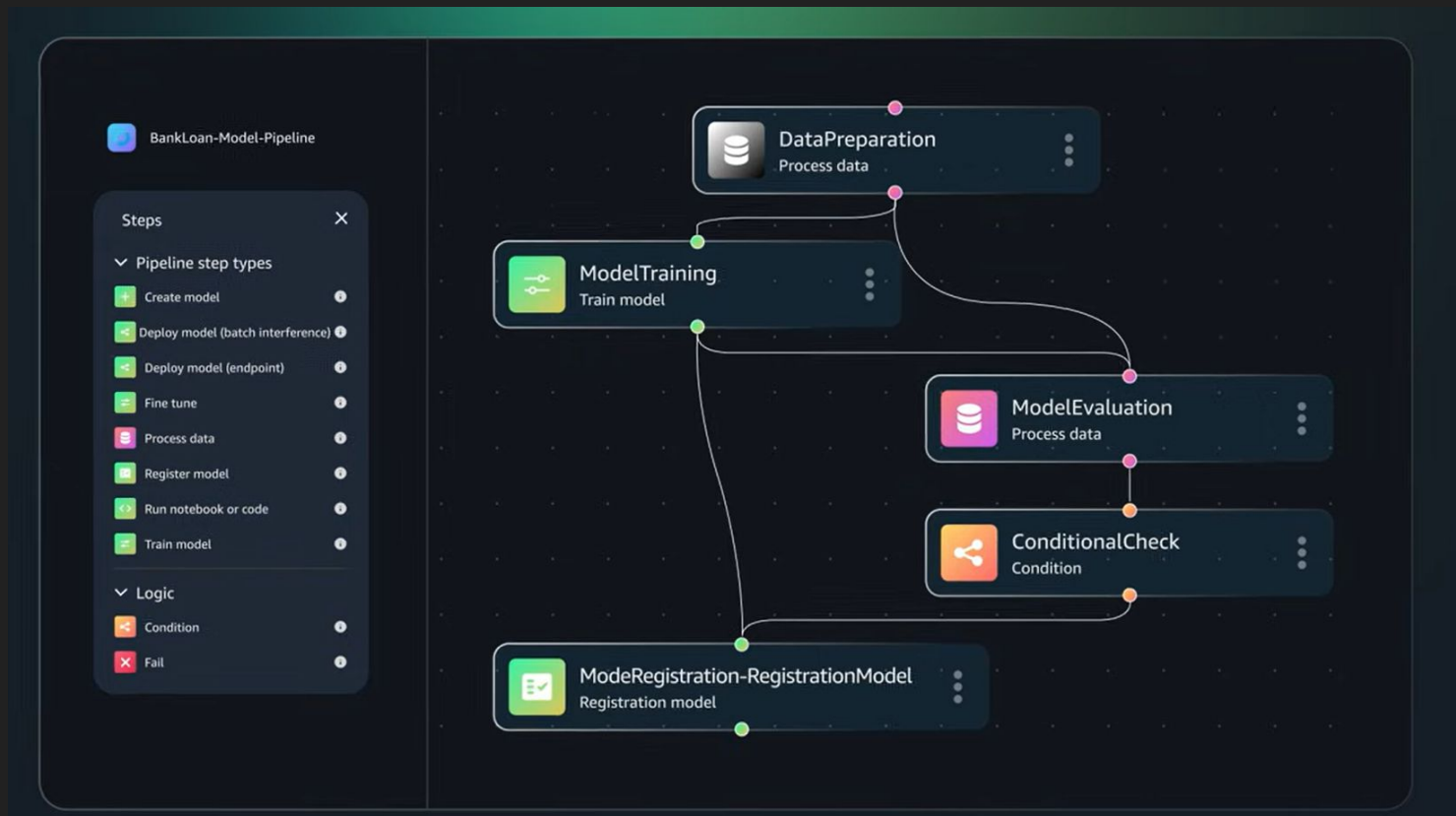
- What's New with Amazon



4. Tutorial: Monitoring Dashboard



4. Tutorial: Pipeline Automatization



4. Tutorial: MLflow



Apache MLflow

Code Editor

MLflow

Canvas

Partner AI Apps

Home

Running Instances

Compute

Data

AutoML

Experiments

Jobs

Pipelines

Models

JumpStart

Deployments

MLflow

Tracking Servers

Create and manage your tracking servers

Create

Genai-agents-v3

Created

Genai-agents-v3

Created

How it works

Create an MLflow Tracking Server

Get started with ML experimentation by creating an MLflow Tracking Server

Create tracking server

Log MLflow experiments

Log metrics, parameters, and model information to view and compare experiments

Open Notebook

Register MLflow models

Register MLflow Models in Model Registry to manage and deploy models that are candidates for production

Open Notebook

MLflow

RAGEvaluation

Overview

Model metrics

System metrics

Artifacts

Run Name

Created

Status

Run ID

Duration

Parent run

Dataset used

Tags

Source

Logged models

Registered models

Parameter

Value

Metric

Value

MLflow

Experiments

Search metrics charts

HPODemo-2024-1...

fearless-zebra...

skittish-carp-83

nosy-elk-711

secretive-loon...

serious-tern-980

unique-sheep...

HPODemo-2024-1...

lyrical-zebra-27

vaunted-boar...

treasured-flea-7

magnificent-wa...

salty-ant-697

MNIST

Rename

Delete

Compare

Search metric charts

loss

test_accuracy

test_average_loss



5. Pricing Model

Pay **only for what you use**, billed per second.



What you pay for

- Compute
- Storage
- Data transfer



The Student Gotcha

Endpoints bill **24/7** until **shut down**, even if they are idle.



Free Tier

Includes free usage hours for notebooks, training, and hosting.

6. Conclusions - When to Use SageMaker?



LUNA PROJECT INSIGHTS

SageMaker is best for **scaling complex AI workflows**.



Seamless Scalability

Effortlessly handles growth in data volume, hospital nodes, and user demand.



Unified Ecosystem

Integrates data prep, training, and monitoring into one cohesive platform.



Hybrid Deployment

Mixes Real-Time chatbots with async VLM jobs for maximum resource efficiency.

Key Takeaway: Use SageMaker when moving from experimentation to production-grade AI projects.



7. Bibliography & Resources

- AWS Sagemaker AI [Website](#)
 - AWS Academy: Machine Learning Foundation Course Documentation.
 - AWS Youtube Channel: [Amazon SageMaker Technical Deep Dive Series \(15 video tutorials\)](#)
 - Medium: [End-to-end ML model deployment using AWS SageMaker](#): Project review by Pratish Mashankar
 - SageMaker Python SDK [Documentation](#)
-
- Resources: CCDBA LUNA Project SageMaker [Tutorial](#)